

**Problem Set 10**  
**Comp 301**  
**Spring 2025**  
**09.05.2025 - 11.05.2025**

**Instructions:**

- Submit your answers to the Blackboard PS10 assignment until May 11th Sunday, at 23.59.
- Please use the code boilerplate, which includes several tests for you to see if your code is correct.
- Save your code and pdf file, zip it as ID\_username.zip with your ID and username (Example: 1234567\_odiler19.zip), and submit this ZIP file.
- Read the questions carefully. Good luck!

**Problem 1:** Evaluate the following expressions. Explain whether they produce the same outcome or not, and why that is the case.

. a)

```
1
2 let a = newref(10) in
3   let g = let b = newref(deref(a)) in
4     proc(c)
5       begin
6         setref(c, -(deref(b), 1))
7         setref(b, -(deref(b), -1))
8         deref(c)
9       end
10   in let x = (g a) in
11     let y = (g a) in
12       -(x, y)
```

. b)

```
1
2 let a = newref(10) in
3   let g = proc(c)
4     let b = newref(deref(a)) in
5       begin
6         setref(c, -(deref(b), 1))
7         setref(b, -(deref(b), -1))
8         deref(c)
9       end
10   in let x = (g a) in
11     let y = (g a) in
12       -(x, y)
```

**Problem 2:** For the code provided in Problem 1(a), draw the environment and store diagrams when value-of is used for the expression deref (c) (line 8) first time. You can use the diagram templates provided below. You can leave some boxes empty, or add additional boxes if necessary.

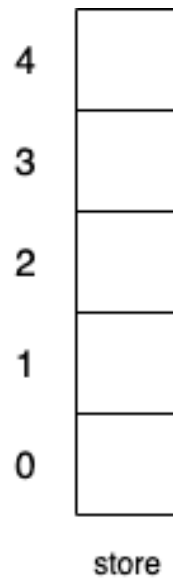
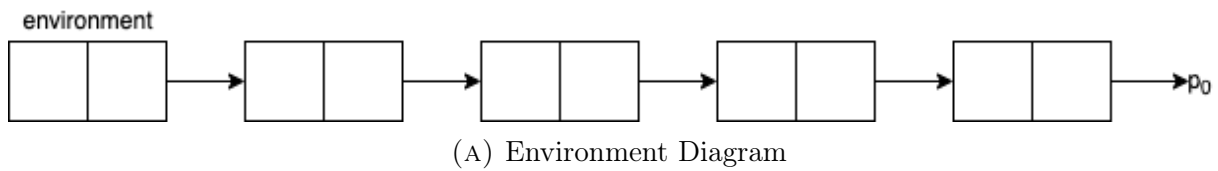


FIGURE 1. Diagrams to use for Problem 2

**Problem 3:** Modify the `interp.scm` so that a `setref-exp` returns the old contents of the location.

**Problem 4:** <sup>1</sup>: Implement the store in constant time by representing it as a Scheme vector. You can look at the documentation of Scheme vectors.

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<sup>1</sup>EOPL Exercise 4.9

Solutions Problem Set 10  
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**Problem 1:** First one evaluates to (num-val -1) Second one evaluates to (num-val 1)  
They give different results because in the second one, parameter b is local to the procedure body, or in other words it is re-evaluated whenever the procedure is called.

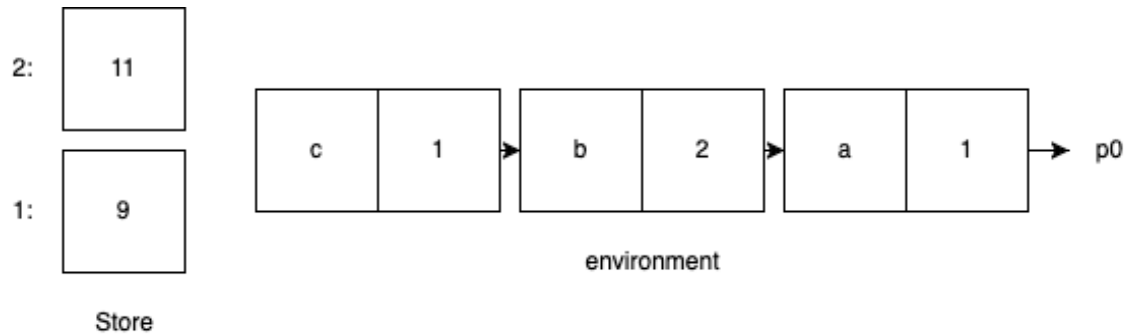


FIGURE 1. problem 2 solution

**Correction:** The values in the environment should be ref-val(1), ref-val(2), and ref-val(1) instead of numbers.